SeaAlert

Classifying Maritime Distress Messages
with LLM - Generated Data

Presenter:

Tomer Atia, Ilana Estrin

Project Review-SeaAlert - Interim

Goal: Classify maritime VHF messages into 4 severity levels (Routine, Safety, Urgency, Distress)

Input & Output

- **input**: message text (synthetic VHF-style transcript)
- **Output**: 4-class severity label

What changed from the proposal

- Evaluate on a hard subset: actual_codeword = NONE
- Evaluate with masking: replace codewords with [CODEWORD]

Novelty / contributions

- Built a balanced synthetic dataset (300 samples, 75 per class) with realistic metadata (style, scenario, etc.)
- Showed that codewords and style can act as shortcuts and measured the drop under masking
- Compared a classical baseline (TF-IDF+LR) vs a transformer baseline (DistilBERT) under the same robust settings

Related Work (Previous Work)

Title	Year	Task	Data & modality	Method	Key results	Relation to SeaAlert
AI-based approach for transcribing and classifying unstructured emergency call data: A methodological proposal. [reference]	2023	ASR + classify emergency calls	2019 "1-9-2" calls (SAMU Brazil)	Speech-to-text + text classification pipeline	Works despite noisy audio	Same pipeline idea: comms -> transcript -> severity
Adaptation and Optimization of ASR for the Maritime Domain in VHF Communication. [reference]	2025	Improve VHF ASR (noise, accents, jargon)	~62h VHF (EN+DE), ~6h real test	Fine-tune XLSR; compare to Whisper (WER)	Lower WER after adaptation	Better transcripts improve downstream severity model
Synthetic Data Generation with Large Language Models for Text Classification: Potential and Limitations. [reference]	2023	When synthetic text helps classifiers	Multi-setting synthetic-text studies	LLM generates labeled text; train classifiers	Helps sometimes, sensitive to ambiguity/eval	Supports our approach + warns to use robust eval (NONE-only, masking)

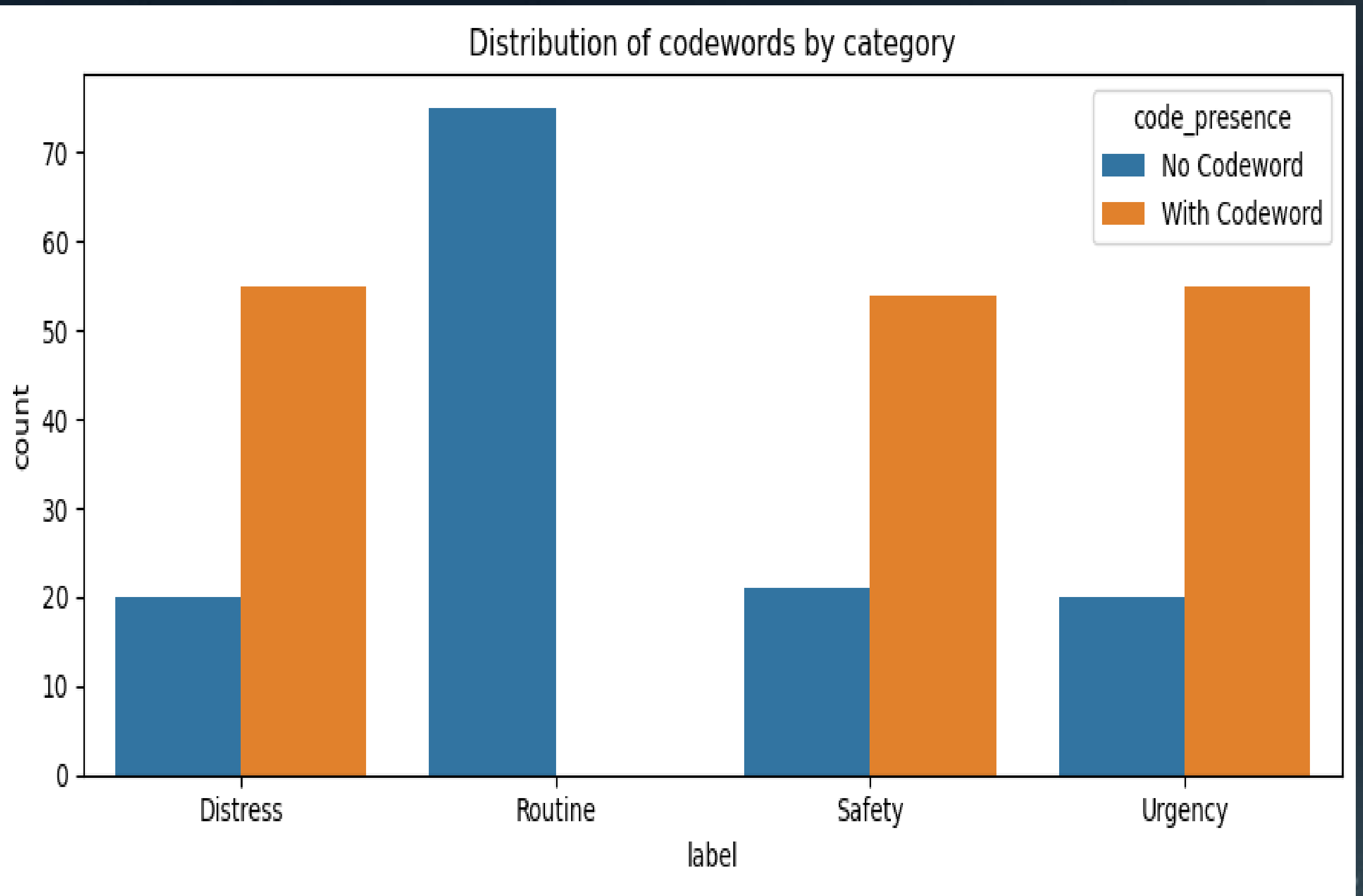
Examples:

Label	Informal (No Codeword)	Third Party	Formal
Distress	"Red Sea Runner,we're taking on water, it's coming in fast and we can't keep up. Visibility's terrible in this fog. Need immediate assistance, anyone nearby respond."	"MAYDAY RELAY, MAYDAY RELAY, MAYDAY RELAY. I'm a nearby boat, I just picked up a MAYDAY from SEA BREEZE, call sign 4XCD2. They report a person overboard off Palmahim Beach and the visibility is really poor – they're struggling to keep sight of them. Any vessel in the area assist immediately. OVER	"MAYDAY, MAYDAY, MAYDAY THIS IS GALILEE STAR, GALILEE STAR, GALILEE STAR CALL SIGN 4XJK5 MMSI 520511037 POSITION 32°28.0'N 034°52.0'E (HADERA AREA) FIRE ON BOARD, STRONG WIND REQUIRE IMMEDIATE ASSISTANCE 3 PERSONS ON BOARD OVER."

Dataset + EDA (300 samples)

Dataset (what we created)

- **Synthetic VHF/SAR message dataset:** 300 samples
- **4 severity labels:** Routine, Safety, Urgency, Distress
- **Balanced:** 75 per class
- **Fields:** text, label, scenario_type, style, vessel, location, weather, POB, codeword



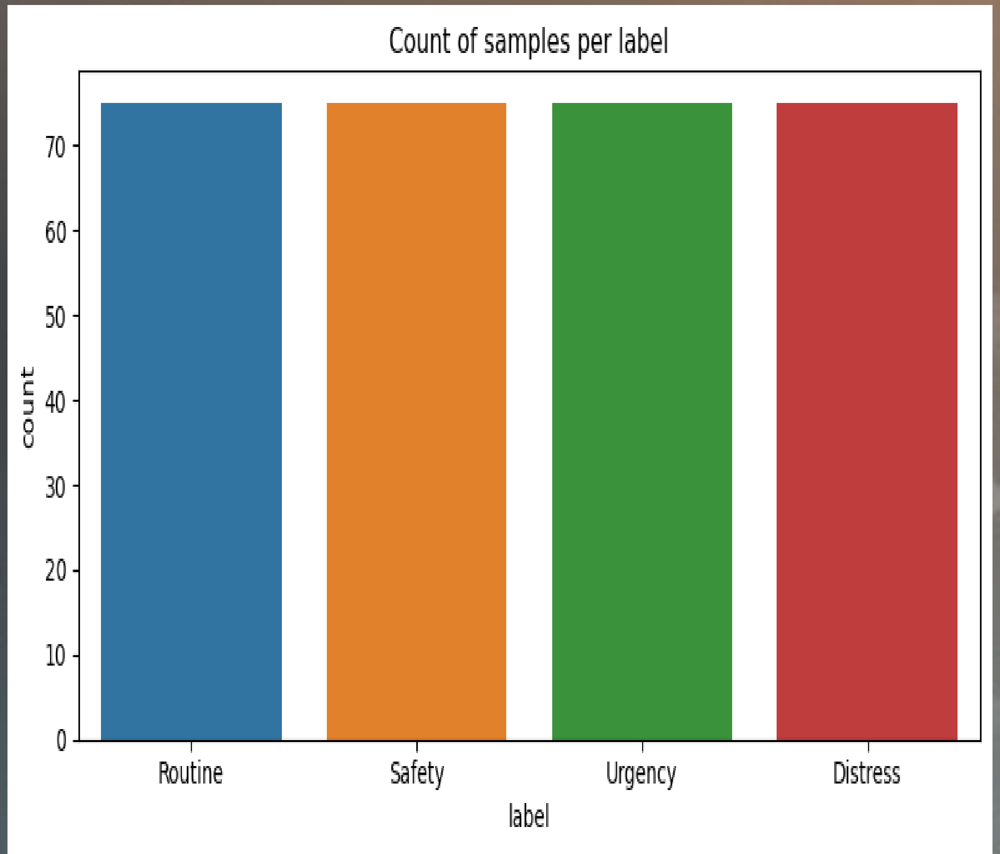
Data generation + labeling

- Messages were **LLM-generated** with controlled scenario + metadata (style, vessel, location, weather, POB)
- Labels assigned by **scenario severity** (4-class target)
- **Codewords** may appear (MAYDAY / PAN PAN / SECURITE) to reflect real radio protocol
- Added **codeword extraction + masking** to support robustness tests in Stage 2

EDA highlights

- **Clean core fields:** 0 missing in text+label, no duplicates
- **Balanced classes:** 75 per class
- **Shortcut signals:** codewords present in ~55% + style-label correlation

Takeaway: evaluate on NONE-only + masked input to test true context understanding



Baseline Solution and Results

Baselines

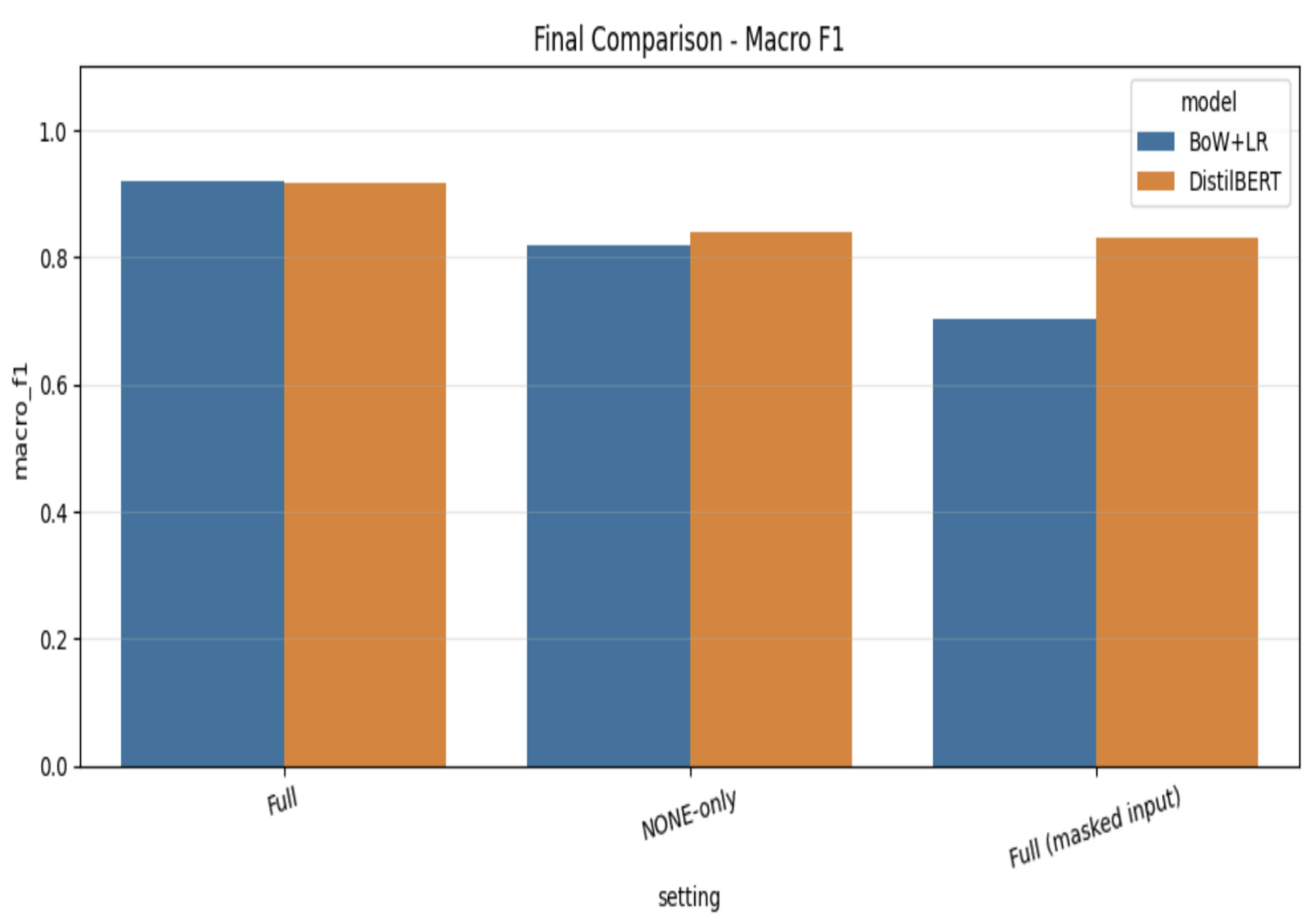
- **Model A (classical):** TF-IDF + Logistic Regression
- **Model B (pretrained):** DistilBERT fine-tuning, 4-class head

Results

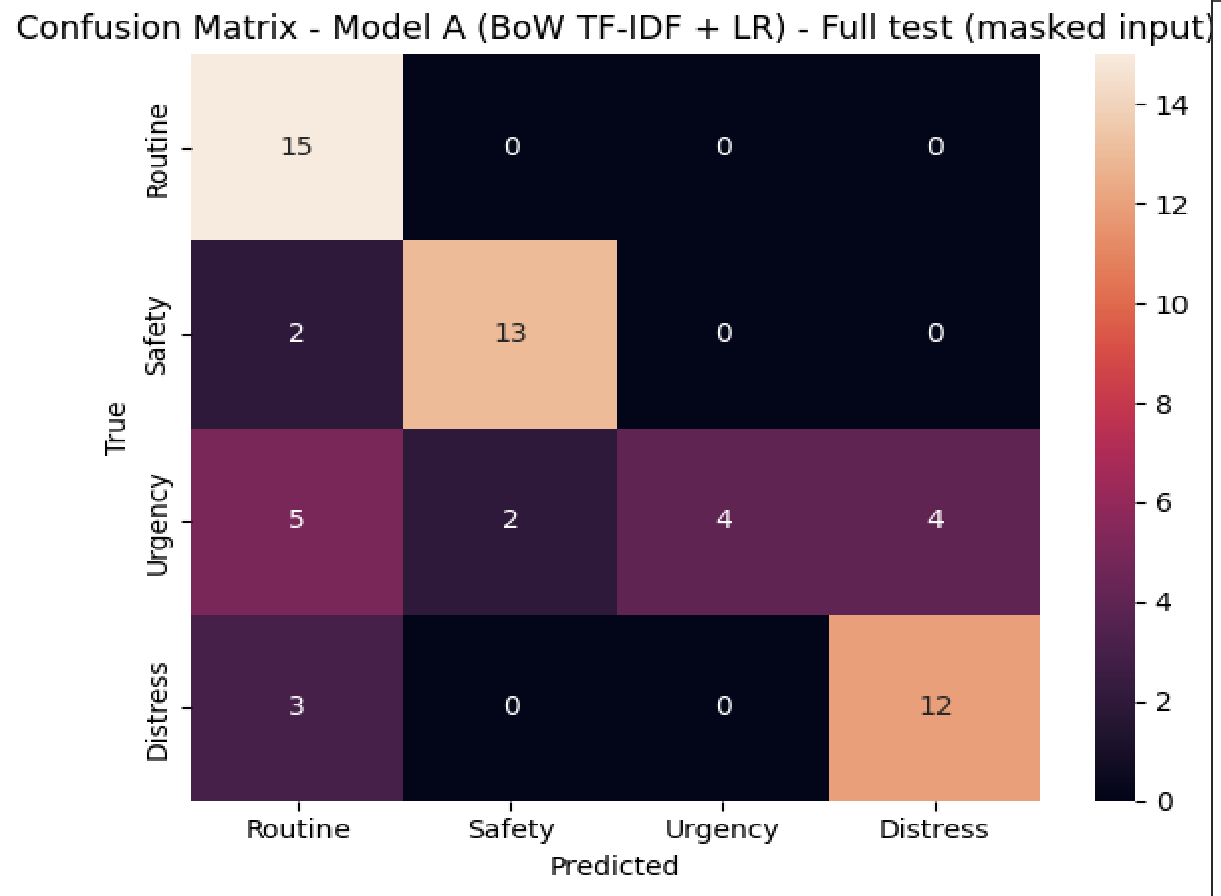
- **Key finding:** Masking causes a large drop (especially BoW+LR) → indicates shortcut reliance on codewords.
- **DistilBERT is more robust** under masking and performs best on NONE-only.

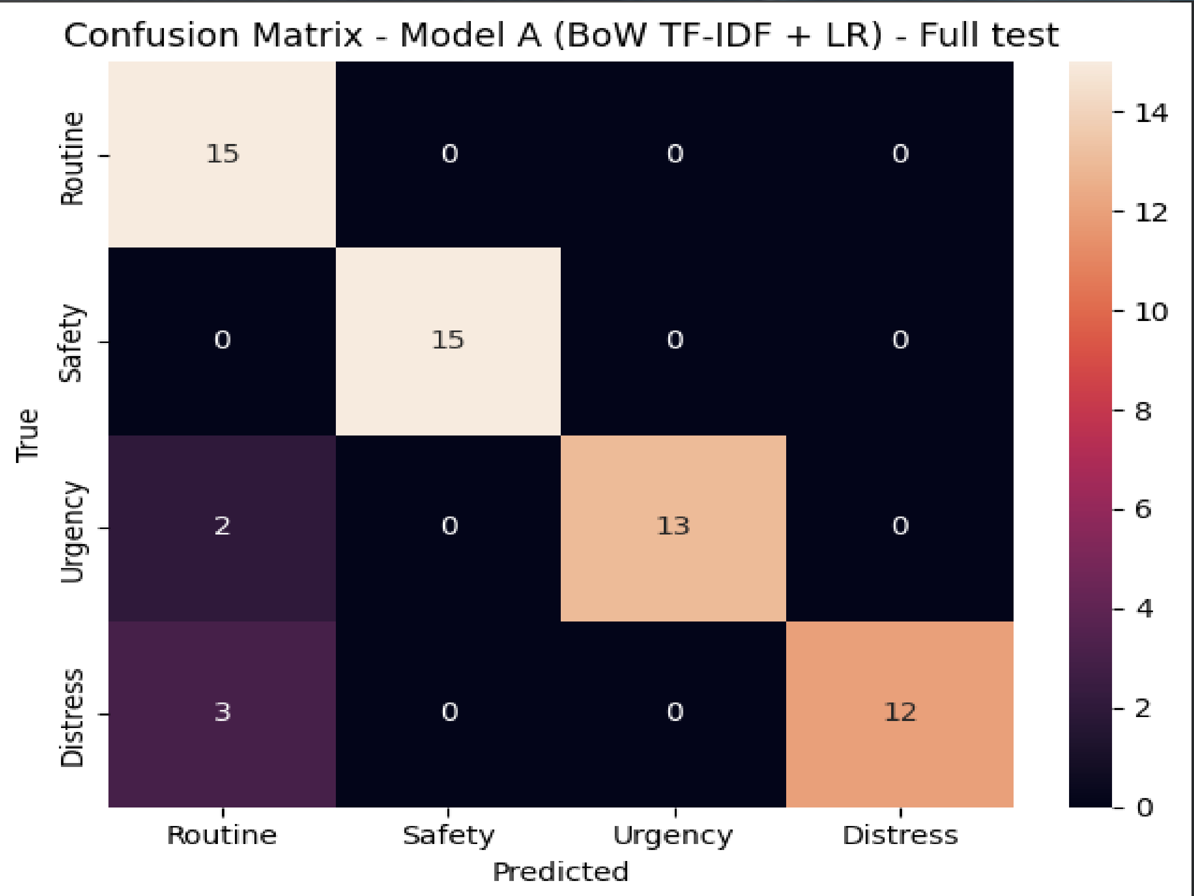
Evaluation setup

- **Full test:** standard split (60 samples)
- **NONE-only:** test samples with actual_codeword = NONE (31 samples)
- **Masked input:** replace MAYDAY / PAN PAN / SECURITE with **[CODEWORD]**



BoW+LR Confusion Matrices (Full vs Masked):





Plan (Roadmap to Final Project)

Step	Scope (what)	Technical (how)	Due date	Expected outcome
1	Expand dataset to **1,000–1,500** samples	Generate new scenarios, keep **balanced classes**, keep metadata (style, vessel, weather, location, POB)	Week 1	Larger, more diverse dataset, fewer artifacts
2	Improve dataset quality	Add **hard cases** (implicit distress), add **borderline** examples, reduce “template” patterns	Week 1	Better realism, harder evaluation
3	Stronger baselines	Fine-tune **RoBERTa-base** and/or **DeBERTa-v3-base** for 4-class classification	Week 2	Higher Macro-F1, better generalization
4	Robustness evaluation	Re-run **Full / NONE-only / Masked**	Week 2	Proof model learns context
5	Error analysis	Inspect confusion pairs + analyze failures (esp. Urgency vs Distress), sample-level review	Week 2	Clear explanation of weaknesses + next fixes
6	Optional: LLM baseline	Add **LLM zero-shot** (prompted classification) as reference baseline	Week 3	Compare classical vs transformer vs LLM
7	Package + final deliverables	Clean notebook, tables/graphs, conclusions, limitations	Week 3	Ready report + reproducible results
8	**Prepare final presentation**	Final slides, final figures, key takeaways + demo examples	Final week	Final presentation ready